INCH POUND

MIL-R-50659B (AR)
28 March 1990
SUPERSEDING
MIL-R-50659A (AR)
30 October 1979

MILITARY SPECIFICATION

REGULATOR, 23-VOLT, BOARD ASSEMBLY: 10549014

This specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 23-volt regulator board assembly, hereinafter referred to as a 23-volt regulator, used in the regulator assemblies of Power Supply PP-4812, part of Radar Set AN/VPS-2 and in the Subassembly PP-6324, part of Radar Test Set AN/TPM-22 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-45743 - Soldering, Manual Type, High Reliability Electrical and Electronic Equipment
MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the (Military Specifications and Standards Bldg. 4D 700 Robbins Avenue Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548823 - Inductor, Power
10548830 - Transformer, Saturable Core
10549014 - 23-Volt Regulator Board Assembly

PACKAGING DATA SHEET

PS10549014 - Packaging of 23-Volt Regulator

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10549014, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the 23-volt regulator shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The 23-volt regulator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The 23-volt regulator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The 23-volt regulator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

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3.3.3.2 Non-operating. The 23-volt regulator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The 23-volt regulator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg. F and +155 deg. F.

3.3.4.2 Non-operating (storage). The 23-volt regulator shall not suffer permanent damage when subjected to the constant temperature extremes of -60 °F for a period of 24 hours and +160 °F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The 23-volt regulator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the 23-volt regulator board assembly.

3.5 Performance. The 23-volt regulator shall produce results as given in 3.5.1 when supplied with the input voltages specified in Table I (Input characteristics).

TABLE I. Input characteristics.

REQUIREMENT	CHARACTERISTIC
Input voltage 1/	26 to 38 Vdc; regulated to ±0.5%
Input signal 1/	22 ± 1 KHz, 71.5 ± 12.5 V peak-to-peak square wave

1/ Input voltages and signals are connected into the circuit using test fixture as specified in 4.6.2.3 and test setup as specified in 4.7.3.

3.5.1 Output terminals. The output terminals are E13 (+) and E14 (-).

3.5.1.1 Output voltage level potentiometer R25. R25 shall control the output voltage between circuit board terminals E13 (+)

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and E14 (-). The nominal output voltage shall be 23.00 ± 0.10 volts (V), with a load current of 1.2 ± 0.2 ampere direct current (Adc).

3.5.1.2 Current limit potentiometer R40. R40 shall control the point at which the current limiting circuitry becomes effective. With a short applied between E13 and E14, maximum current drain shall be controlled by R40 and shall not exceed 1.8 A. The output shall automatically recover after removal of the short circuit.

3.5.1.3 Voltage regulation, line. The output voltage shall not vary more than ±0.23 volts direct current (Vdc) with input voltage variations from 26 to 38 Vdc, and a constant load current of 1.2 A.

3.5.1.4 Voltage regulation, load. The output voltage shall not vary more than ±0.23 Vdc with the input voltage fixed at 32 ± 6 Vdc when the output current varies from 0 to 1.2 A.

3.5.1.5 Output ripple. The ripple voltage shall not exceed 0.23 V peak-to-peak when the input voltage is 32 ± 0.32 Vdc and the load current is 1.2 A. Exclude high-frequency spikes which occur at approximately 25 usec intervals.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The workmanship and soldering requirements of MIL-F-13926, MIL-S-45743 and MIL-S-46844 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature +50 deg. F to +90 deg. F
b. Relative humidity 90% RH max.
c. Pressure 28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) 23-volt regulators for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II (Functional tests) and Table III (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any assembly, component or test specimen to comply with any of the stated requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies, and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

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TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One 23-volt regulator out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the requirements and tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling.

CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Signal generator	Capable of producing a 22 ± 1 KHz, 71.5 ± 12.5 V peak-to-peak square wave. Output impedance less than 20 ohms.
2.	Power supply	DC, 26 to 38 Vdc at 2 A, regulated to 0.5%.
3.	Voltmeter	DC, 0 to 40 V, 0.01% accuracy, 20,000 ohms/volt or greater.

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TABLE IV. Test equipment list (continued).

ITEM NUMBER	DESCRIPTION
4. Oscilloscope	Input sensitivity: 20 mV/cm. Input impedance: 1 megohm. Vertical passband: DC to 1 MHz. Horizontal deflection: 5 us/cm.
5. Ammeter	DC, 0 to 2 A, 1% accuracy.
6. Test fixture	As specified in Figure 1. See Table V for fixture parts list.

TABLE V. Test fixture parts list.

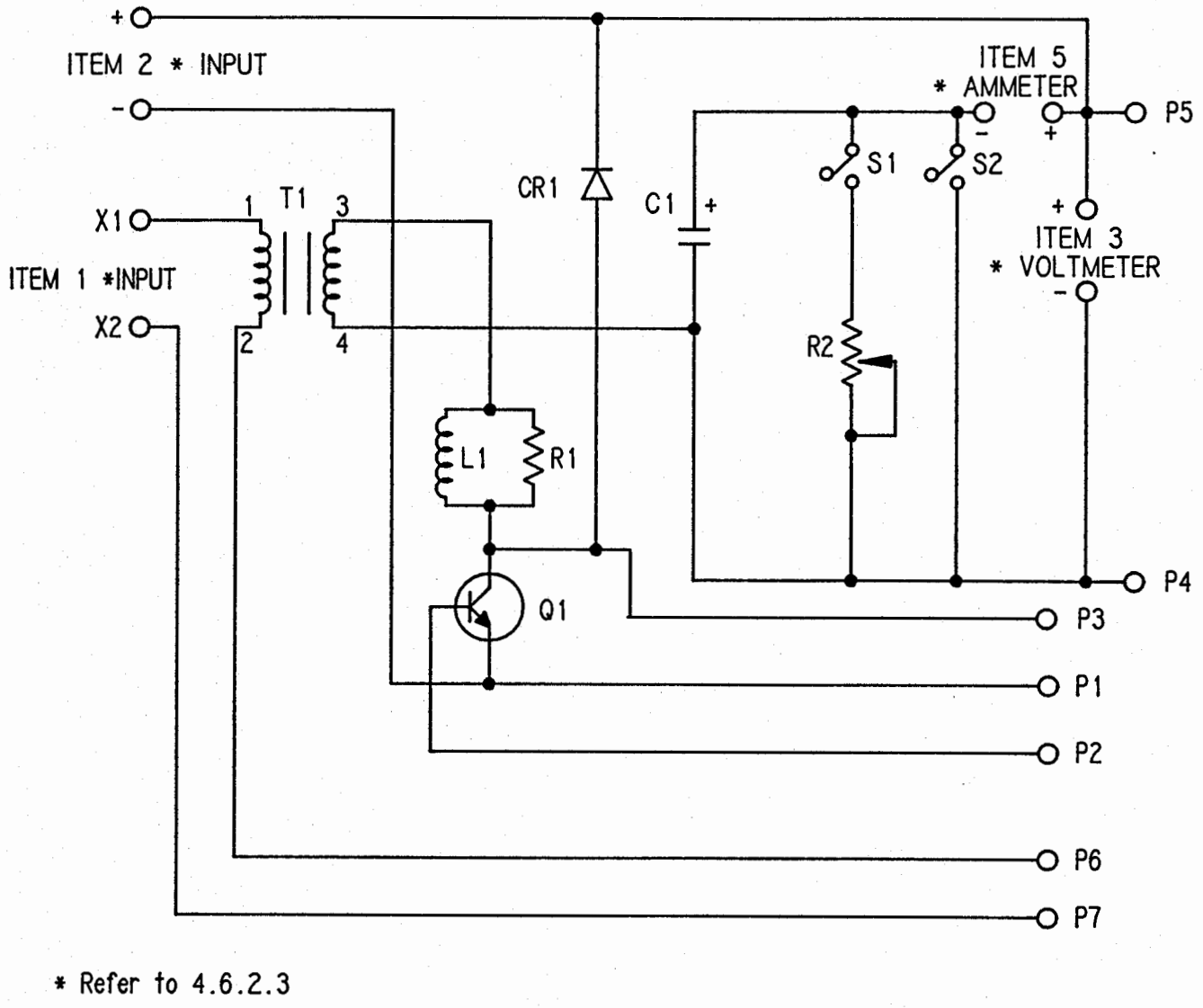
ITEM	NOMENCLATURE	DESCRIPTION
C1	Capacitor	160 uF, 40V
CR1	Diode	General purpose, JAN 1N645
L1	Inductor, power	1.0 mH, 1.5 A, 10548823
Q1	Transistor, power	NPN, silicon, 2N3767
R1	Resistor	Fixed, 2.4 Kohms, 1/2 W, RC20GF242K
R2	Resistor	Variable 0 to 25 ohms, 100 W
S1	Switch	SPST, General purpose, 2 A rated
S2	Switch	SPST, General purpose, 2 A rated
T1	Transformer	Saturable core, 10548830

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The 23-volt regulator shall be attached to the vibration generator directly or with a fixture,

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* Refer to 4.6.2.3

FIGURE 1. Test fixture, regulator board.

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and securely held by its normal means of attachment. After vibration, the 23-volt regulator shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the 23-volt regulator shall be tested as specified in subparagraphs 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500.2 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507.2 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

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4.7.3 Performance. The selected 23-volt regulators shall be tested as follows: All of the tests shall be performed with the test equipment connected to the 23-volt regulator as shown in Figure 2.

4.7.3.1 Output voltage level potentiometer R25. Using the signal generator (item 1, Table IV), apply a 22 ± 1 kilohertz (KHz) square wave to test fixture terminals X1 and X2 and adjust item 1 so that the amplitude of the input signal is 71.5 ± 12.5 V peak-to-peak. Using power supply (item 2, Table IV), apply 32 ± 6 Vdc between the (+) and (-) terminals on the test fixture. With switch S1 open and using voltmeter (item 3, Table IV) verify that the voltage between P4 (-) and P5 (+) on the test fixture can be adjusted to 23.00 ± 0.10 Vdc.

4.7.3.2 Current limit potentiometer R40. Perform the following steps in sequence:

a. Set up the conditions specified in 4.7.3.1.

b. Adjust R25 for voltage indication of 23.00 ± 0.10 Vdc.

c. Adjust R40 fully CCW.

d. Close load switch S1 (see Figure 1).

e. Close short switch S2 (see Figure 1).

f. Adjust R40 CW until output current equals 1.8 Adc.

g. Open S2.

h. Vary resistive load (R2) to obtain current indication of 1.2 Adc.

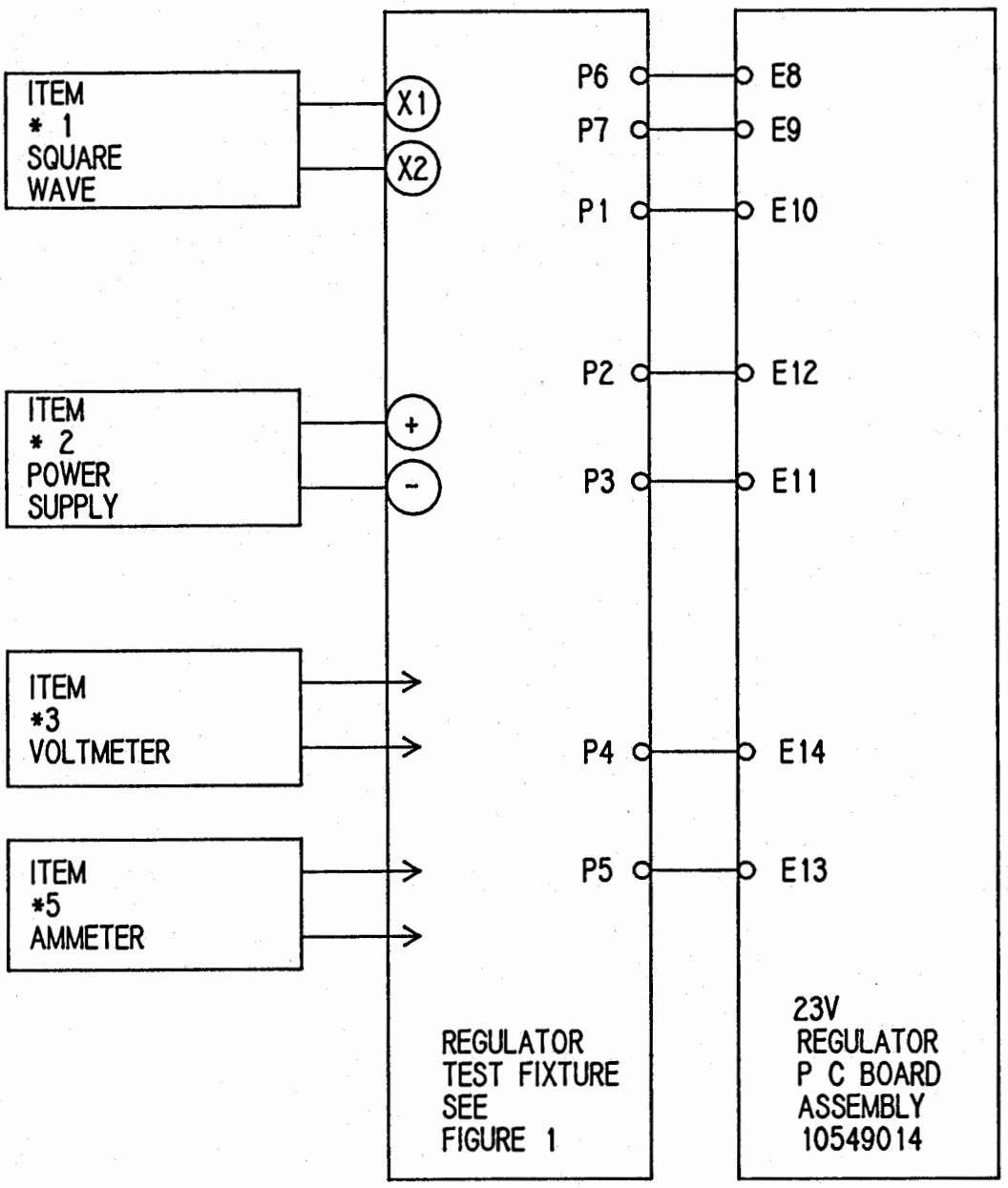
i. Record voltmeter indication using item 3.

j. Close S2 momentarily, then open.

k. Using item 3, verify that output automatically returns to within ±0.10V of the level in "i" above.

4.7.3.3 Voltage regulation, line. With R25 and R40 adjusted as in 4.7.3.2, adjust the resistive load (R2) to obtain 1.2 A. Vary the input voltage DC (item 2, Table IV) from 26 to 38 Vdc. Verify that the output voltage does not vary more than ±0.23 Vdc.

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* Refer to 4.6.2.3

FIGURE 2. Test harness setup.

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4.7.3.4 Voltage regulation, load. Set the input voltage to 32 ± 6 Vdc. With R25 and R40 adjusted as in 4.7.3.2, adjust the resistive load (R2) to maximum and open switch S1. Record voltage indication using voltmeter (item 3, Table IV). Close switch S1 and vary the resistive load to obtain 1.2 A. Using voltmeter (item 3, Table IV), verify that the output voltage does not vary more than ±0.23 Vdc.

4.7.3.5 Output ripple. With the input voltage set at 32 ± 0.32 Vdc and R25 and R40 adjusted as in 4.7.3.2, adjust the resistive load (R2) to 1.2 A. Using oscilloscope (item 4, Table IV), verify that the ripple does not exceed 0.23 V peak-to-peak. Exclude high-frequency spikes which occur at approximately 25 usec intervals.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P10549014.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 23-volt regulator is used in the regulator assemblies of the Power Supply PP-4812, part of Radar Set AN/VPS-2. It is also used in Subassembly PP-6324, part of Radar Test Set AN/TPM-22. The 23-volt regulator is used to regulate the output voltage within the specified requirements for functional operation of Radar Set AN/VPS-2. In the test set, it is used to check voltage regulator performance of the radar power supply.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.

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6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire control system
Power Supply PP-4812/VPS-2
Radar Set AN/VPS-2
Radar Test Set AN/TPM-22
Regulator, 23-volt
Subassembly PP-6324/TPM-22
Vulcan Air Defense Systerm (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A686)

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